later_seen | equal-weighted | Absolute loss surfaces
Expanding controls are deduplicated sidebands, not finite coordinates
Exploratory seen-period estimates; calibration/power deferred; no calibrated significance claim

Separate | A: matched PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	1.016	0.933	0.895	0.885	0.879	0.874	0.865	0.853	0.849	0.853	0.861	0.865	0.866	0.865	0.866	0.867	0.867	0.867	0.867
7	0.994	0.924	0.896	0.879	0.873	0.862	0.861	0.856	0.846	0.849	0.858	0.860	0.863	0.862	0.864	0.867	0.864	0.863	0.864
6	0.983	0.918	0.884	0.881	0.866	0.859	0.859	0.849	0.844	0.850	0.859	0.861	0.858	0.858	0.862	0.861	0.859	0.860	0.861
5	0.973	0.926	0.880	0.870	0.859	0.854	0.852	0.846	0.837	0.844	0.850	0.858	0.858	0.855	0.857	0.856	0.857	0.855	0.856
4	0.960	0.920	0.876	0.870	0.861	0.862	0.855	0.836	0.835	0.842	0.857	0.863	0.862	0.859	0.859	0.856	0.859	0.859	0.863
3	0.950	0.928	0.869	0.869	0.859	0.858	0.846	0.841	0.838	0.838	0.853	0.858	0.856	0.854	0.854	0.850	0.848	0.848	0.851
2	0.931	0.920	0.872	0.868	0.850	0.850	0.849	0.837	0.834	0.840	0.850	0.854	0.854	0.853	0.853	0.849	0.848	0.849	0.849
1	0.919	0.913	0.869	0.865	0.856	0.853	0.851	0.843	0.839	0.841	0.854	0.855	0.853	0.854	0.853	0.850	0.849	0.850	0.851

Coefficient-history pairs

Separate | B: expanding PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	1.059	0.971	0.927	0.911	0.901	0.897	0.896	0.891	0.872	0.872	0.882	0.882	0.875	0.868	0.869	0.873	0.870	0.870	0.868
7	1.037	0.959	0.914	0.899	0.888	0.884	0.884	0.883	0.865	0.870	0.881	0.883	0.877	0.871	0.871	0.870	0.866	0.866	0.865
6	1.003	0.943	0.898	0.889	0.880	0.873	0.871	0.873	0.859	0.867	0.878	0.880	0.873	0.867	0.868	0.867	0.864	0.863	0.862
5	0.988	0.937	0.891	0.886	0.875	0.867	0.867	0.871	0.859	0.865	0.875	0.878	0.872	0.868	0.868	0.868	0.865	0.863	0.863
4	0.973	0.936	0.891	0.889	0.876	0.869	0.869	0.867	0.860	0.860	0.875	0.876	0.869	0.864	0.865	0.863	0.860	0.858	0.858
3	0.958	0.931	0.880	0.877	0.864	0.862	0.865	0.861	0.861	0.862	0.878	0.879	0.873	0.869	0.870	0.867	0.865	0.863	0.862
2	0.942	0.929	0.879	0.876	0.860	0.854	0.853	0.848	0.844	0.844	0.853	0.856	0.855	0.853	0.853	0.849	0.848	0.847	0.849
1	0.916	0.913	0.868	0.865	0.856	0.852	0.850	0.843	0.839	0.841	0.854	0.855	0.853	0.853	0.853	0.850	0.849	0.849	0.850

Coefficient-history pairs

Shared control

PC count K	Exp
8	0.848
7	0.848
6	0.843
5	0.844
4	0.839
3	0.843
2	0.841
1	0.837

Coefficient-history pairs

Common | A: matched PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	0.903	0.898	0.865	0.874	0.863	0.862	0.854	0.846	0.845	0.848	0.854	0.858	0.856	0.859	0.862	0.866	0.868	0.866	0.865
7	0.907	0.895	0.866	0.872	0.864	0.860	0.855	0.848	0.845	0.849	0.853	0.856	0.857	0.859	0.862	0.867	0.867	0.866	0.864
6	0.910	0.895	0.865	0.874	0.864	0.859	0.856	0.849	0.845	0.849	0.857	0.857	0.855	0.857	0.862	0.867	0.867	0.867	0.866
5	0.909	0.897	0.865	0.873	0.861	0.857	0.852	0.847	0.839	0.845	0.852	0.855	0.853	0.856	0.860	0.862	0.864	0.863	0.864
4	0.905	0.892	0.861	0.869	0.856	0.859	0.850	0.835	0.833	0.841	0.854	0.858	0.857	0.858	0.862	0.863	0.868	0.869	0.871
3	0.899	0.895	0.857	0.867	0.855	0.854	0.840	0.831	0.830	0.832	0.843	0.844	0.843	0.845	0.846	0.846	0.846	0.847	0.849
2	0.893	0.892	0.860	0.865	0.846	0.848	0.846	0.835	0.831	0.837	0.847	0.847	0.847	0.849	0.849	0.848	0.849	0.849	0.849
1	0.894	0.892	0.861	0.854	0.847	0.846	0.844	0.838	0.833	0.836	0.847	0.847	0.846	0.848	0.848	0.846	0.846	0.846	0.847

Coefficient-history pairs

Common | B: expanding PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	0.925	0.908	0.871	0.873	0.864	0.861	0.861	0.859	0.848	0.853	0.862	0.864	0.862	0.864	0.867	0.871	0.870	0.867	0.865
7	0.911	0.900	0.866	0.867	0.861	0.856	0.855	0.854	0.845	0.851	0.863	0.864	0.863	0.866	0.868	0.869	0.866	0.865	0.864
6	0.903	0.900	0.865	0.867	0.859	0.854	0.851	0.851	0.843	0.850	0.861	0.863	0.862	0.864	0.866	0.867	0.864	0.864	0.863
5	0.901	0.897	0.862	0.866	0.857	0.852	0.851	0.851	0.843	0.848	0.859	0.862	0.859	0.861	0.863	0.863	0.862	0.862	0.862
4	0.901	0.899	0.862	0.866	0.857	0.851	0.847	0.844	0.838	0.841	0.855	0.855	0.851	0.852	0.854	0.853	0.852	0.850	0.851
3	0.899	0.897	0.862	0.865	0.855	0.854	0.855	0.852	0.851	0.854	0.866	0.867	0.863	0.863	0.864	0.864	0.862	0.861	0.861
2	0.895	0.894	0.858	0.857	0.847	0.844	0.843	0.838	0.834	0.837	0.846	0.847	0.847	0.848	0.849	0.847	0.847	0.847	0.849
1	0.892	0.892	0.860	0.855	0.847	0.846	0.844	0.838	0.834	0.836	0.847	0.847	0.846	0.848	0.848	0.846	0.846	0.846	0.847

Coefficient-history pairs

Shared control

PC count K	Exp
8	0.848
7	0.849
6	0.847
5	0.844
4	0.839
3	0.842
2	0.839
1	0.838

Coefficient-history pairs

Partial | A: matched PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	0.928	0.914	0.882	0.882	0.866	0.861	0.857	0.847	0.842	0.847	0.853	0.856	0.853	0.851	0.857	0.860	0.864	0.859	0.861
7	0.928	0.910	0.884	0.876	0.864	0.859	0.858	0.853	0.848	0.845	0.852	0.852	0.850	0.848	0.857	0.859	0.856	0.857	0.859
6	0.930	0.908	0.877	0.875	0.866	0.858	0.864	0.852	0.847	0.848	0.859	0.856	0.852	0.851	0.856	0.857	0.857	0.861	0.859
5	0.931	0.906	0.878	0.877	0.865	0.857	0.858	0.846	0.838	0.842	0.852	0.856	0.853	0.851	0.855	0.854	0.854	0.855	0.856
4	0.915	0.904	0.873	0.873	0.861	0.859	0.845	0.836	0.834	0.841	0.853	0.858	0.855	0.854	0.856	0.857	0.859	0.858	0.862
3	0.911	0.900	0.872	0.873	0.860	0.858	0.846	0.838	0.833	0.836	0.847	0.850	0.846	0.845	0.846	0.844	0.844	0.844	0.846
2	0.914	0.903	0.881	0.876	0.854	0.853	0.854	0.839	0.835	0.839	0.852	0.852	0.847	0.850	0.849	0.845	0.844	0.845	0.844
1	0.907	0.909	0.869	0.864	0.854	0.854	0.850	0.843	0.839	0.842	0.853	0.852	0.850	0.851	0.851	0.848	0.848	0.848	0.848

Coefficient-history pairs

Partial | B: expanding PCA

PC count K	52	78	104	130	156	182	208	234	260	286	312	338	364	390	416	442	468	494	520
8	0.973	0.917	0.882	0.887	0.868	0.869	0.867	0.866	0.848	0.854	0.861	0.861	0.856	0.858	0.863	0.863	0.863	0.862	0.862
7	0.947	0.910	0.881	0.879	0.865	0.861	0.858	0.859	0.843	0.851	0.860	0.861	0.857	0.858	0.862	0.863	0.858	0.861	0.858
6	0.911	0.904	0.879	0.879	0.863	0.859	0.853	0.850	0.840	0.849	0.860	0.859	0.853	0.856	0.859	0.858	0.856	0.855	0.854
5	0.912	0.902	0.874	0.876	0.861	0.854	0.848	0.852	0.840	0.849	0.861	0.863	0.858	0.858	0.860	0.860	0.860	0.858	0.857
4	0.922	0.904	0.880	0.879	0.862	0.855	0.847	0.848	0.838	0.843	0.857	0.856	0.851	0.852	0.855	0.854	0.853	0.852	0.851
3	0.928	0.910	0.880	0.881	0.867	0.863	0.861	0.853	0.849	0.851	0.868	0.868	0.861	0.861	0.864	0.859	0.860	0.861	0.859
2	0.912	0.912	0.871	0.868	0.857	0.855	0.855	0.844	0.841	0.841	0.851	0.854	0.852	0.851	0.851	0.846	0.846	0.846	0.848
1	0.906	0.909	0.868	0.864	0.854	0.854	0.850	0.843	0.839	0.842	0.852	0.852	0.850	0.850	0.850	0.848	0.847	0.847	0.848

Coefficient-history pairs

Shared control

PC count K	Exp
8	0.847
7	0.843
6	0.841
5	0.842
4	0.837
3	0.842
2	0.838
1	0.837

Coefficient-history pairs

Mean target MSE (squared log units)

K=0 own-history baseline: one reference per L, shared across modes/rules

Own control